Colliding Asteroids

AlgoExpert · Medium · Stacks

The problem

- Sign is direction (+ right, - left), magnitude is size.
- When they meet the smaller explodes (both if equal).
- Return survivors.

Intuition

- Simulate collisions with a stack of surviving asteroids.

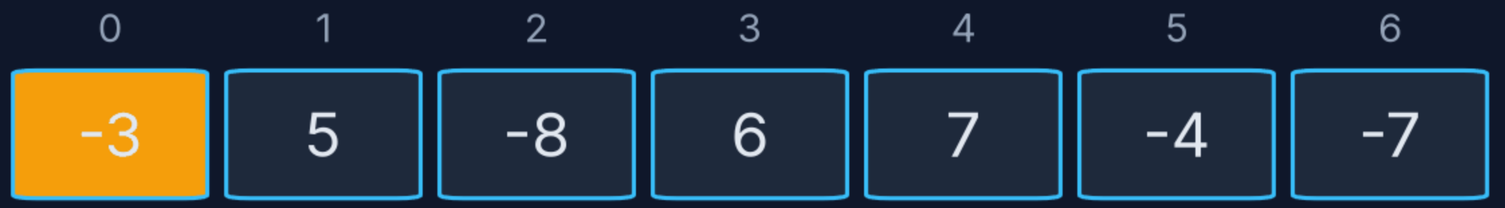
Brute force vs optimal

- Simulate repeatedly until stable → $O(n^2)$
- Stack of survivors → each pushed/popped once, O(n)
- Collisions only: right-mover then left-mover

Approach

- 1. Right movers push
- 2. A left mover destroys smaller right movers on top
- 3. Equal sizes destroy both
- 4. Survivors remain

Worked example



survivors (stack)




```
no collision
```

-3: no collision

Worked example



survivors (stack)




```
no collision
```

5: no collision

Worked example



survivors (stack)



`-8 destroys 5`

-8: -8 destroys 5

Worked example



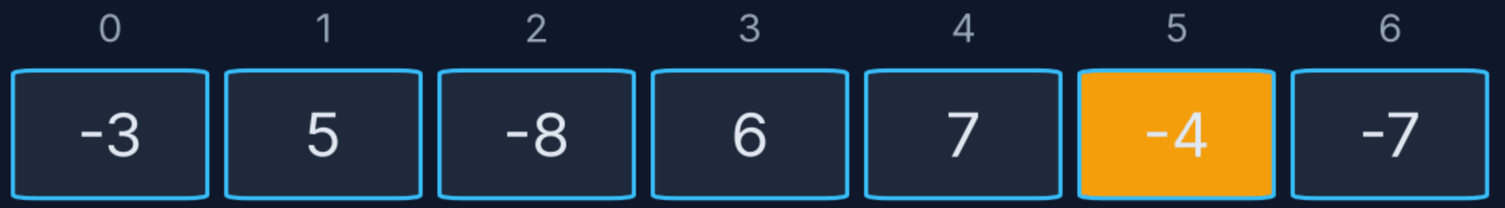
survivors (stack)




```
no collision
```

7: no collision

Worked example



survivors (stack)




```
7 destroys -4
```

-4: 7 destroys -4

Worked example



survivors (stack)




```
-7 and 7 both explode
```

-7: -7 and 7 both explode

Java solution

```java
public class CollidingAsteroids {
    public static int[] collidingAsteroids(int[] asteroids) {
        ArrayDeque<Integer> st = new ArrayDeque<>();
        for (int a : asteroids) {
            boolean alive = true;
            while (alive && a < 0 && !st.isEmpty() && st.peekLast() > 0) {
                int top = st.peekLast();
                if (top < -a) st.pollLast();
                else { if (top == -a) st.pollLast(); alive = false; }
            }
            if (alive) st.addLast(a);
        }
        return st.stream().mapToInt(Integer::intValue).toArray();
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Edge cases & pitfalls

- Left then right never collide
- Equal sizes: both explode
- Negative on empty stack survives

Interview tips & follow-ups

- State the collision condition first

Recap

- Simulate collisions with a stack of surviving asteroids.
- Complexity: O(n)